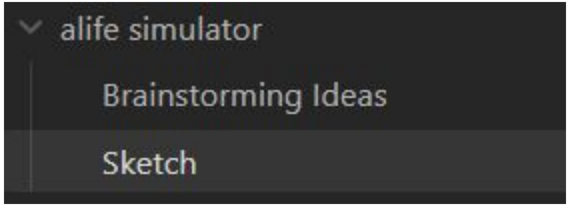
CSSE290: Artificial Life Final Project/Presentation

Final Class before Spring Break

Artificial Life Simulator

Week 5 - ~~you should have selected one and have it approved by instructor~~ You should start exploring the list of artificial life simulators or doing your own search. You may wish to use some of your time over the break to look these without the stress of other deadlines

Week 10 - you will give a presentation on it



You should do the following:

Learn about it and answer questions such as:

- What is the simulator focused on?

- What levels does it attempt to capture?

- What is an example of a research question it purports to be able to help answer?

There are some “games” that may still be perfectly qualified to answer questions. It will be up to you to demonstrate what kind of questions it can answer.

Artificial Life Simulator

Prepare a presentation (~~~10~~ ~25 minutes) that:

- Introduces the simulator

- Explains the components

- Gives an overview of adjustable parameters

- Explains a specific experiment that has been conducted using it

-------------NEW--------------

Report on *INTERESTING/SURPRISING/INSIGHTFUL* results of performing an experiment using it.

I expect you will have to run many experiments to find an interesting one.

You can focus on the most interesting experiment, you don't have to report on all of them.

Artificial Life Simulator - NEW

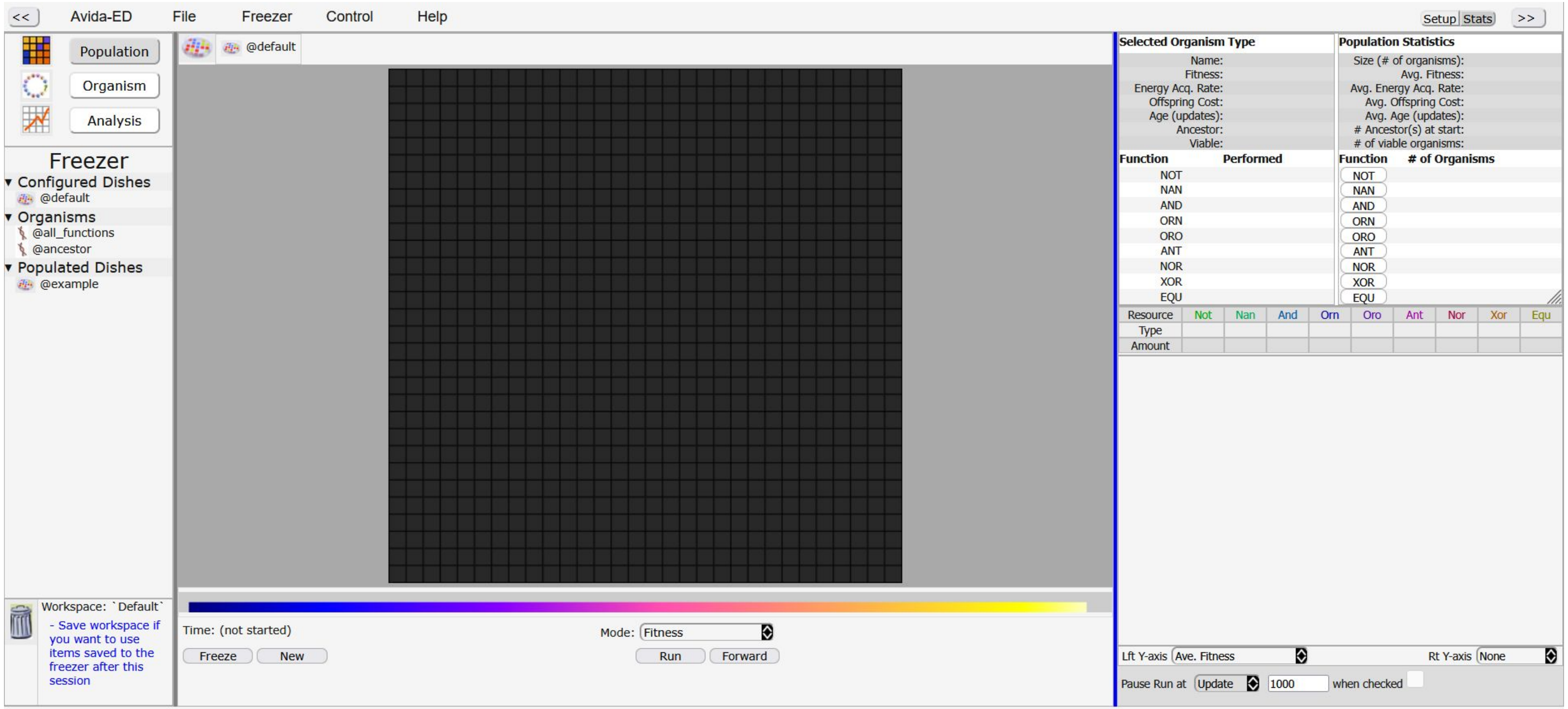
Where applicable it will be best to provide additional analysis/investigation into the results:

- Is it repeatable?
- What are the most important mechanisms (change/remove components)?
- Can we understand in detail one specific instance of the observed results/behavior?
- Could we possibly understand the system better through information theory?
    - How could we apply IT to study this?
- Could we possibly understand the system better through dynamical systems theory?
    - How could we apply DST to study this?
- Is there potential of seeing this behavior/phenomena in the real world?
    - How could you try to imitate this?
    - What would be crucial to consider in evaluating this?

Alex Brickley - AVIDA

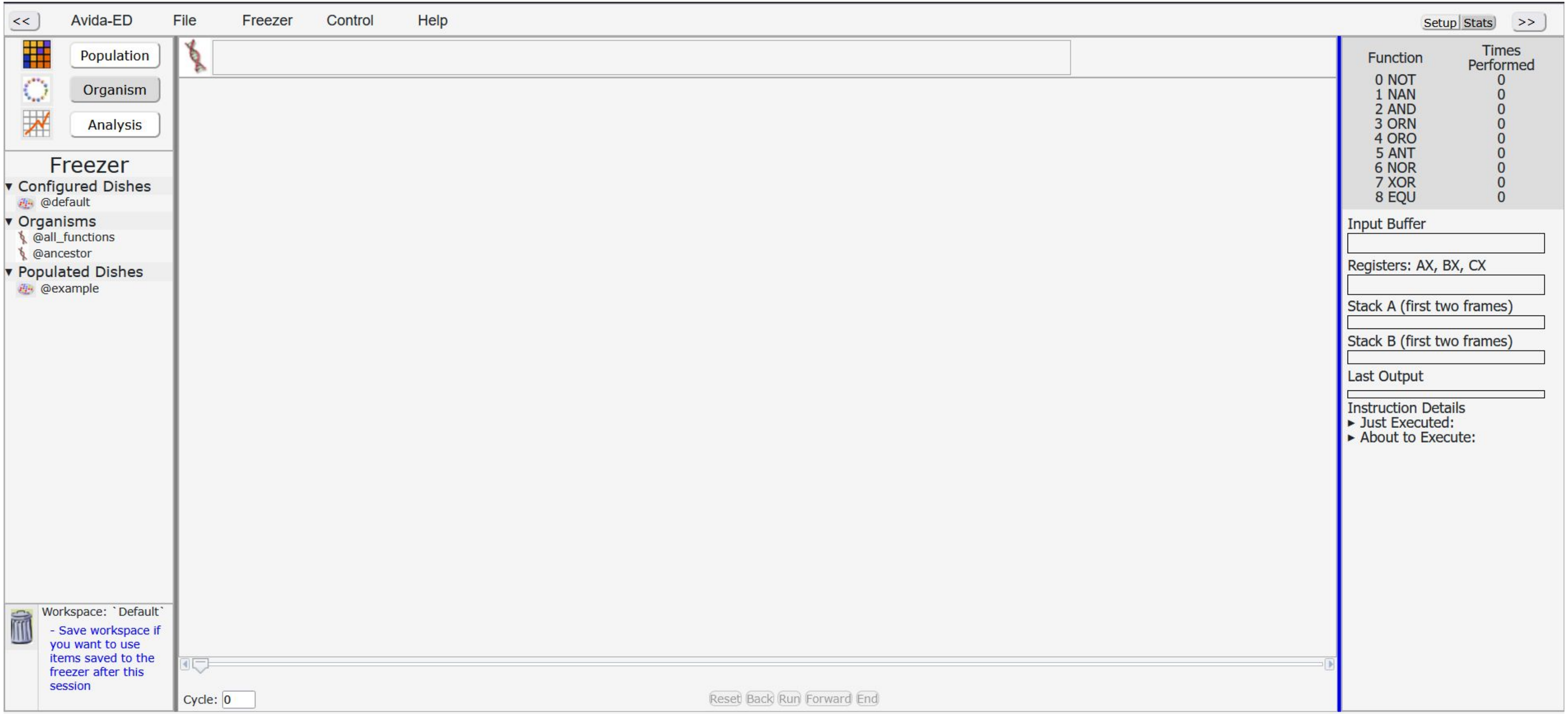
Artificial life platform for research

Population

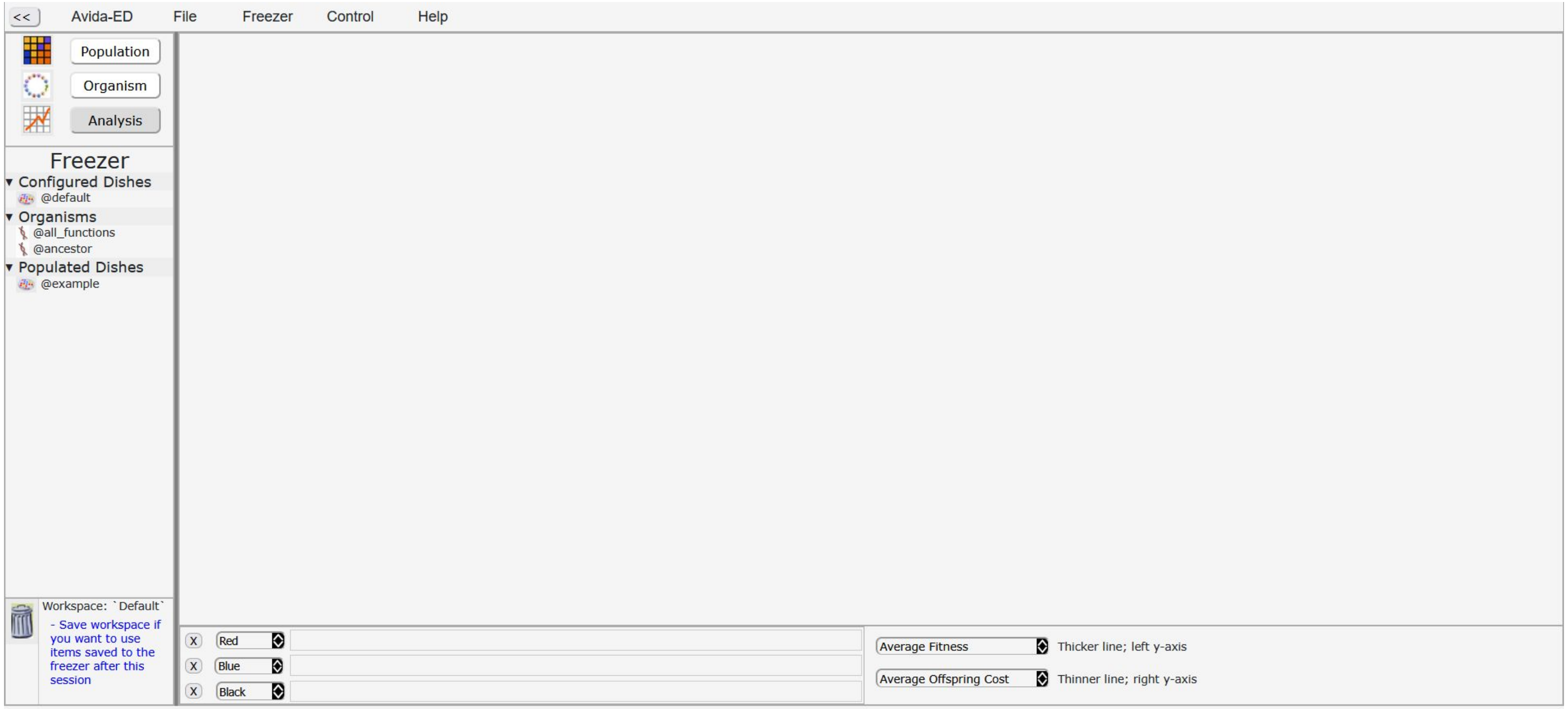


<< Avida-ED File Freezer Control Help
Setup Stats >>
Population
Organism
Analysis
Freezer
Configured Dishes
@default
Organisms
@all_functions
@ancestor
Populated Dishes
@example
Workspace: `Default`
- Save workspace if you want to use items saved to the freezer after this session
@default
Time: (not started)
Mode: Fitness
Freeze New
Run Forward
Selected Organism Type
Name:
Fitness:
Energy Acq. Rate:
Offspring Cost:
Age (updates):
Ancestor:
Viable:
Function Performed
NOT NAN AND ORN ORO ANT NOR XOR EQU
Population Statistics
Size (# of organisms):
Avg. Fitness:
Avg. Energy Acq. Rate:
Avg. Offspring Cost:
Avg. Age (updates):
Ancestor(s) at start:
of viable organisms:
Function # of Organisms
NOT NAN AND ORN ORO ANT NOR XOR EQU
Resource Not Nan And Orn Oro Ant Nor Xor Equ
Type
Amount
Lft Y-axis Ave. Fitness
Rt Y-axis None
Pause Run at Update 1000 when checked

Organism Specific



Analysis Mode



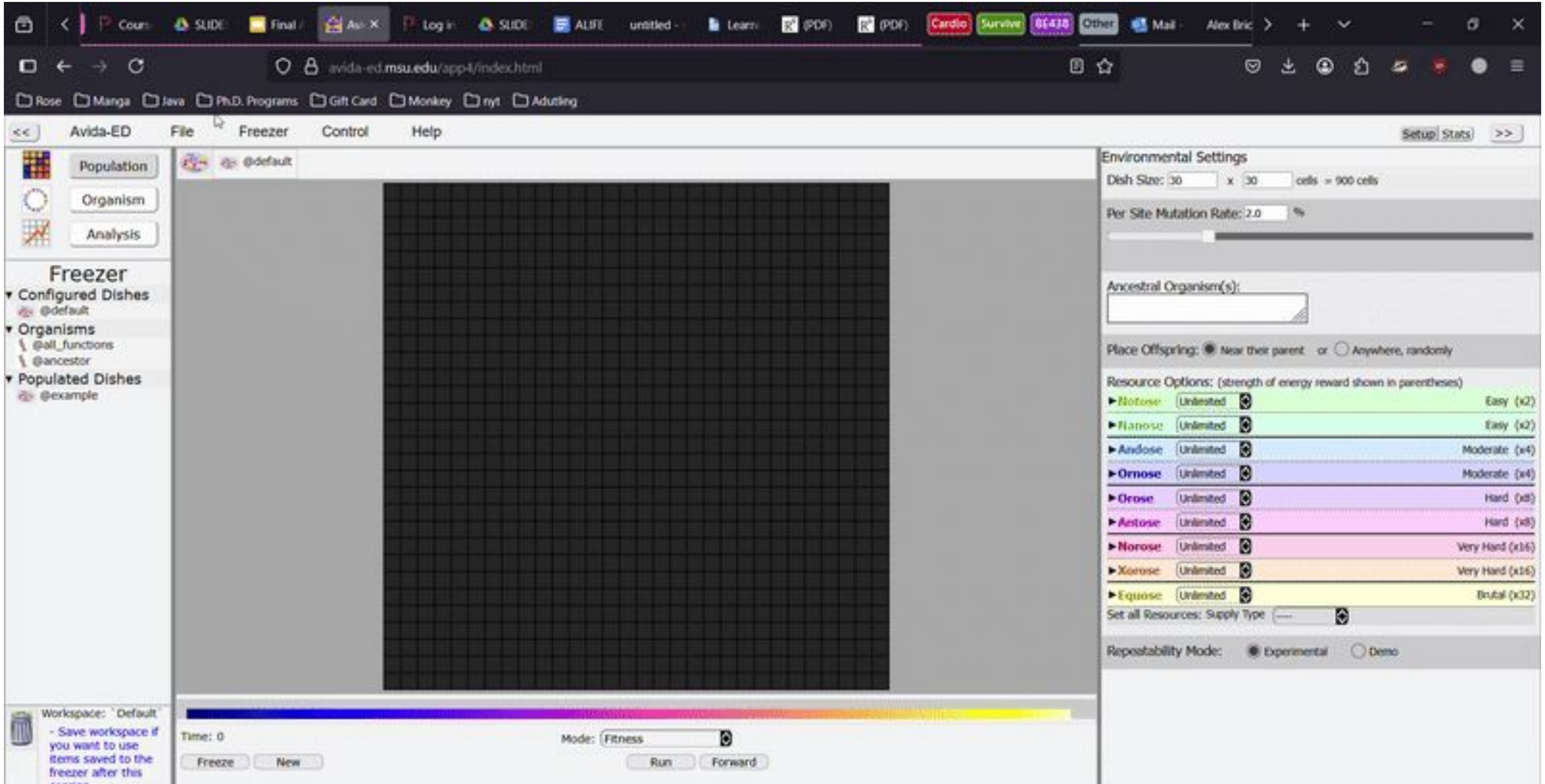
Freezer

Stores experimental setups with specific settings, for replication

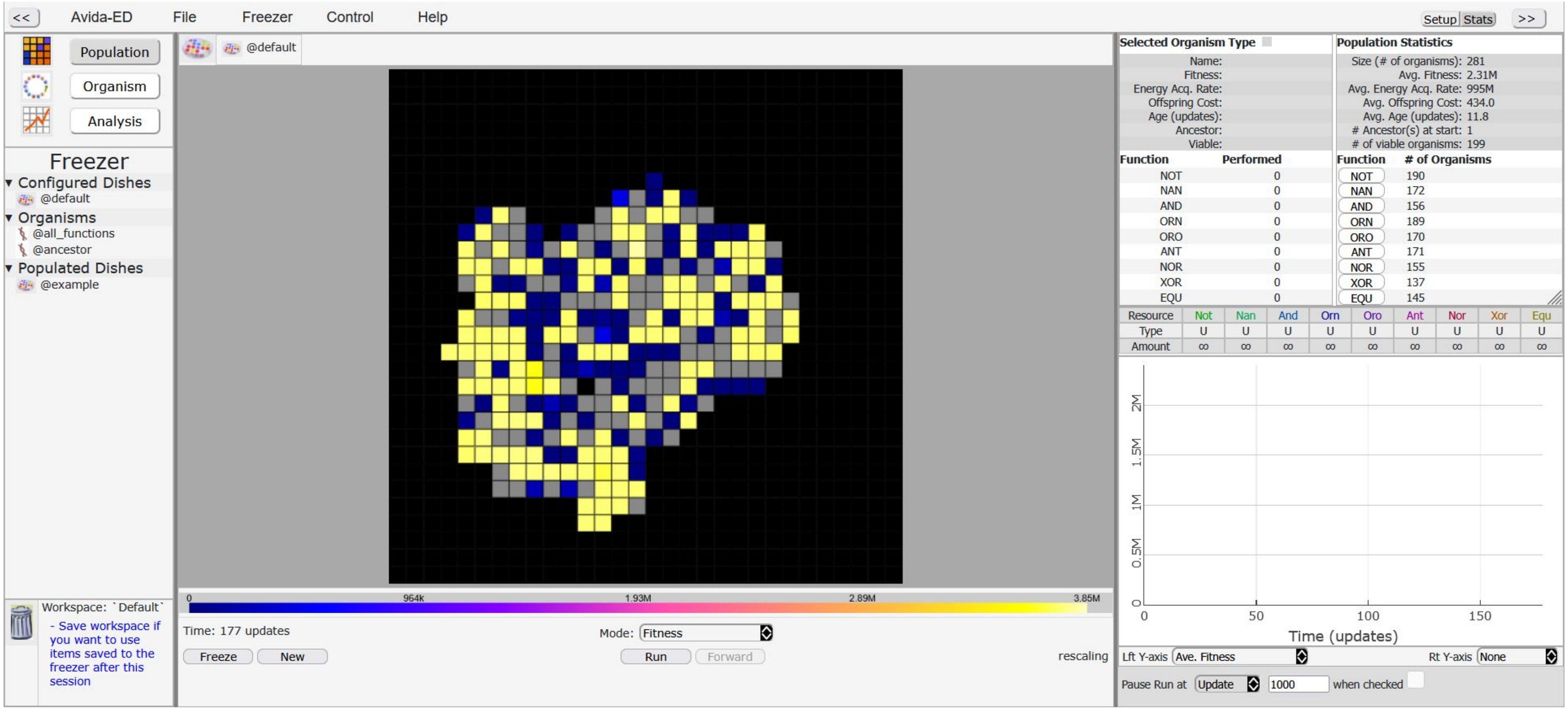
Stores individuals for future use or analysis.

Stores populations for future use or analysis.

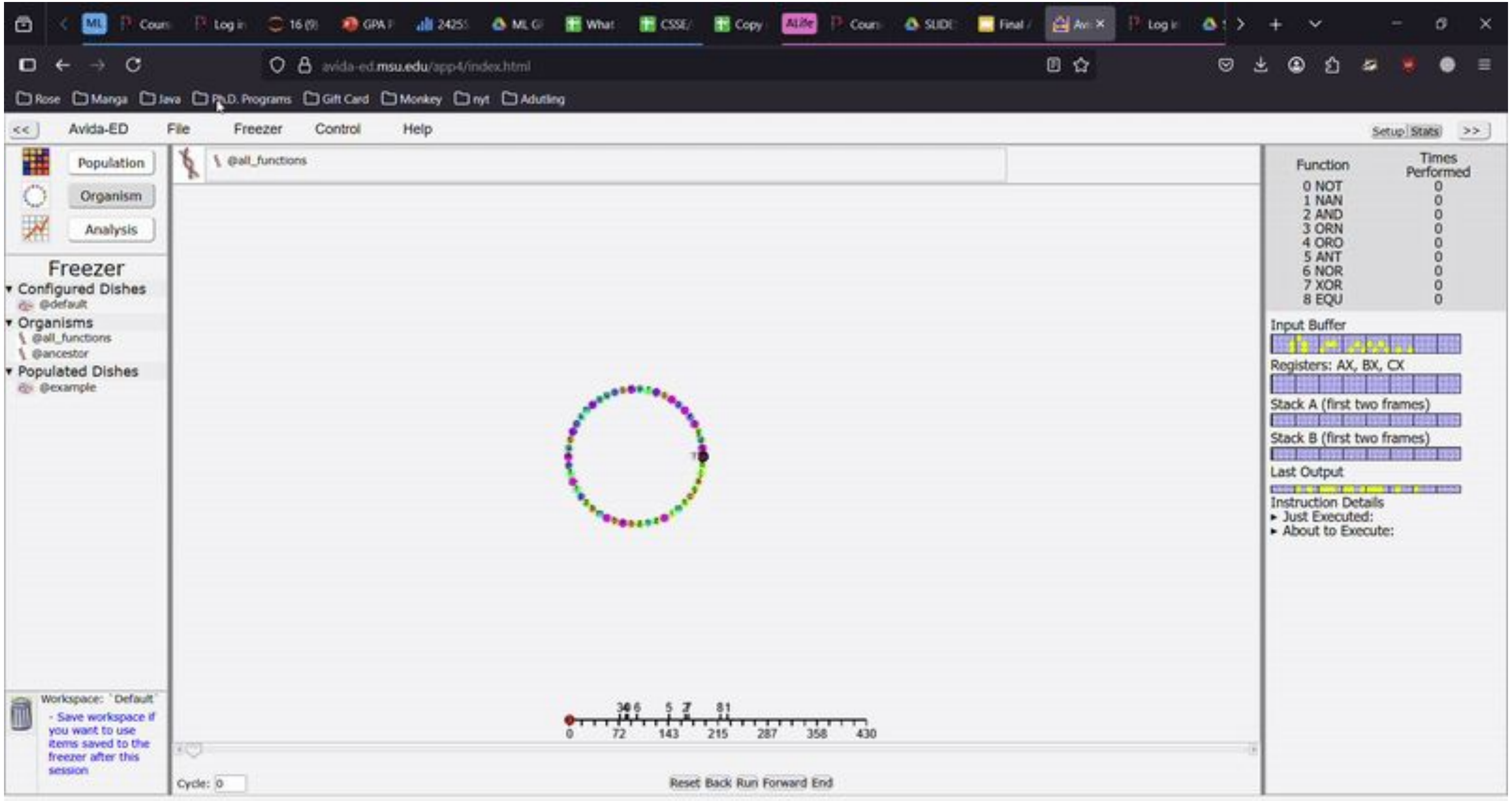
Example population



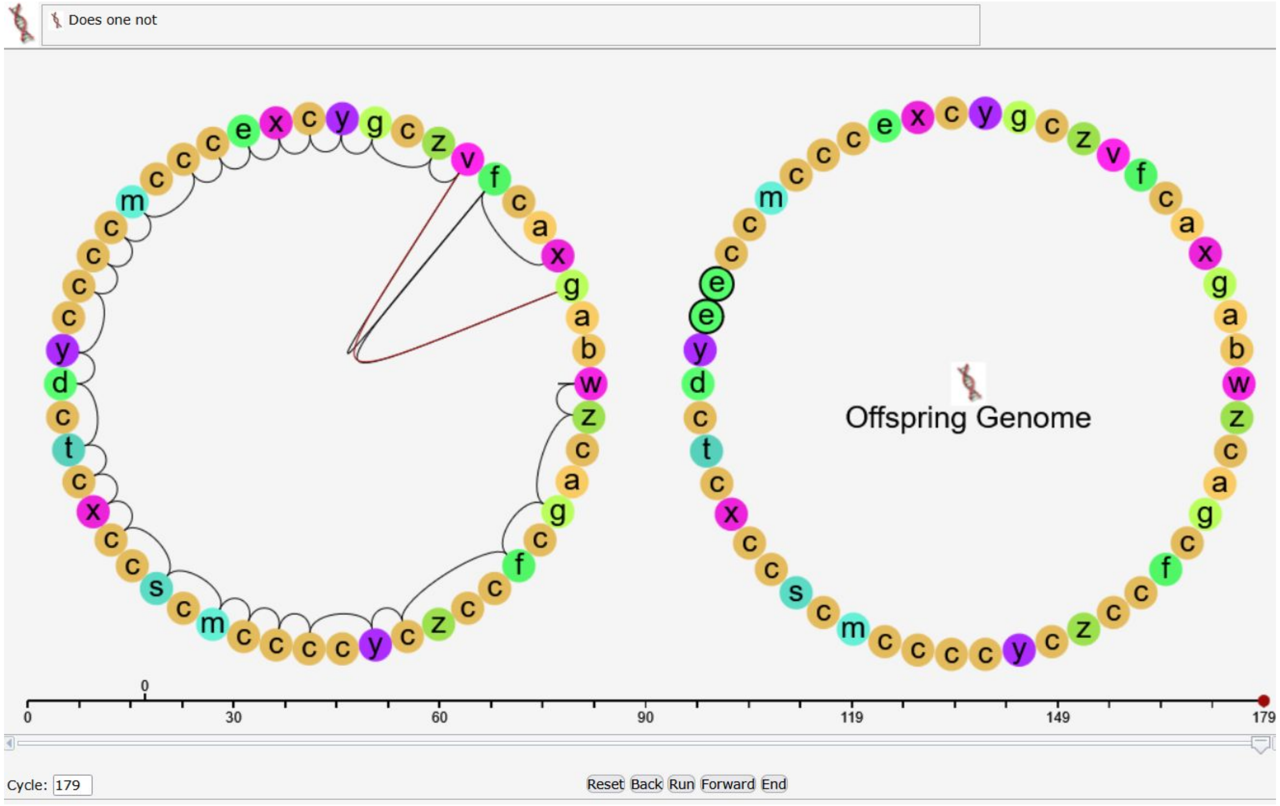
Example population



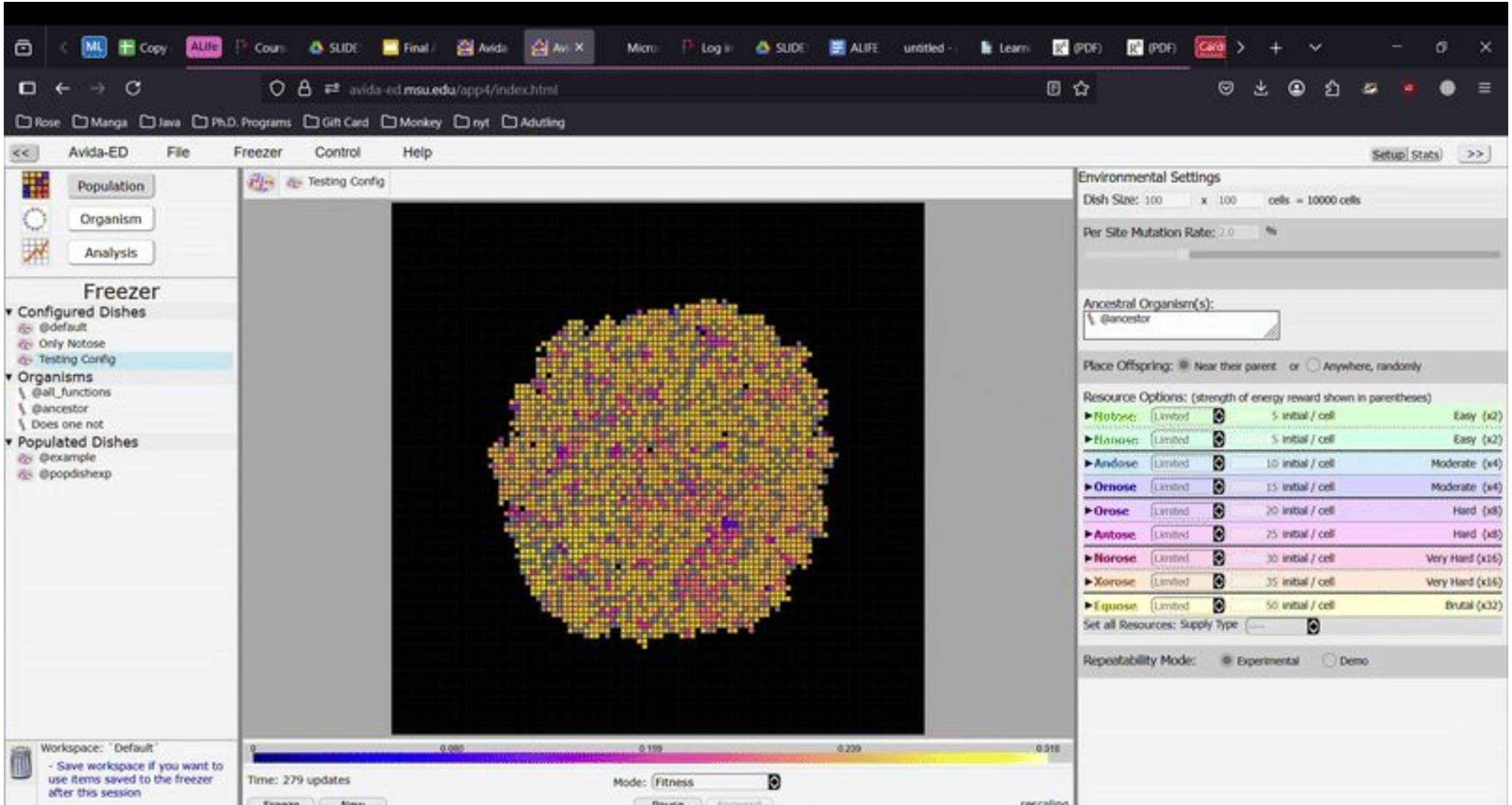
Example of genome replication



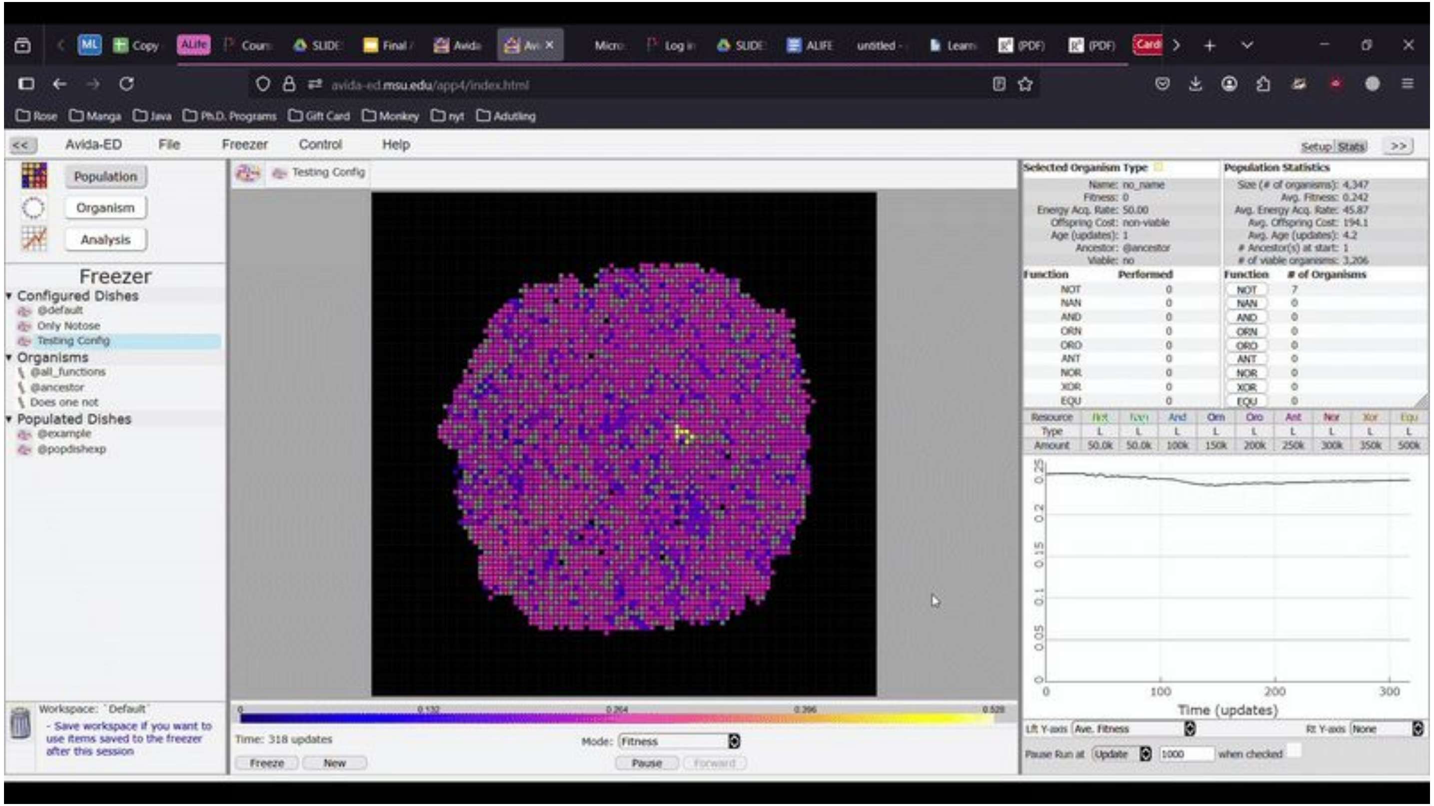
Mutation Notes



Example mutation

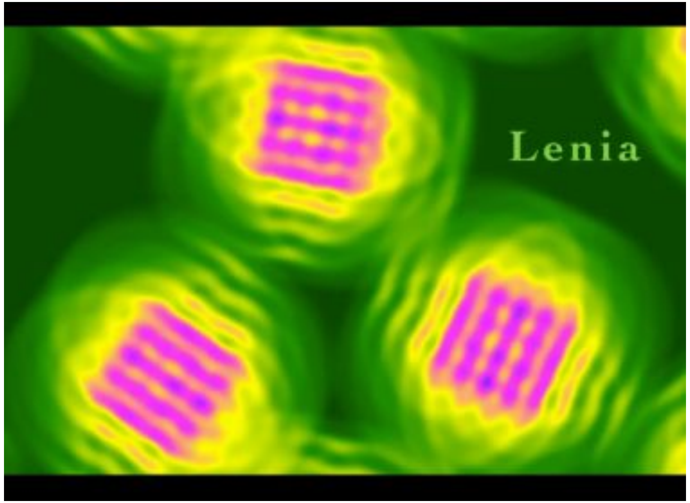


Example mutation



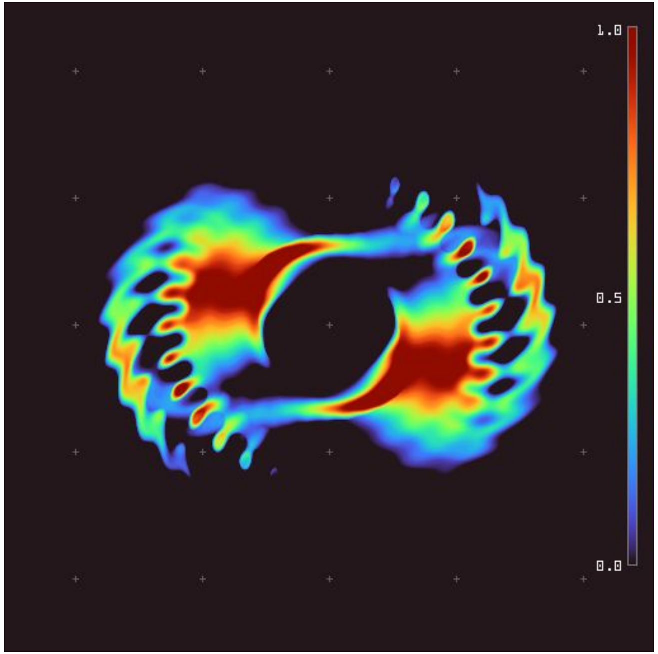
Will Greenlee - Lenia

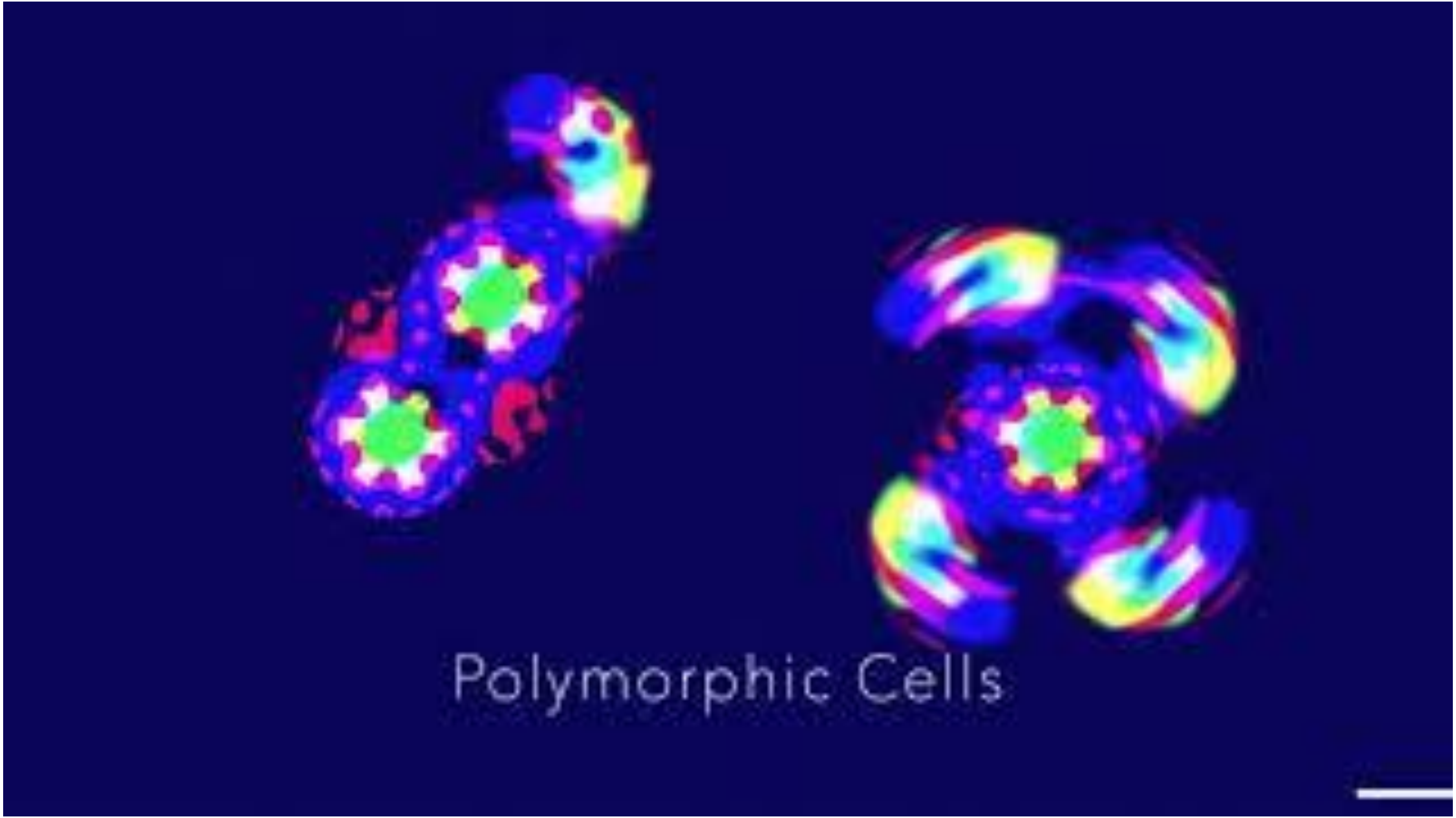
- Created in 2018-19 by Bert Wang-Chak Chan
  - Name from Latin lenis: “smooth”
  - “Computer simulations show that Lenia supports a great diversity of complex autonomous patterns or “life forms” bearing resemblance to real-world microscopic organisms”
  - Written in three major languages
    - Python
    - Matlab
    - Javascript (browser!)
    - https://chakazul.github.io/Lenia/JavaScript/Lenia.html



Will Greenlee - Lenia

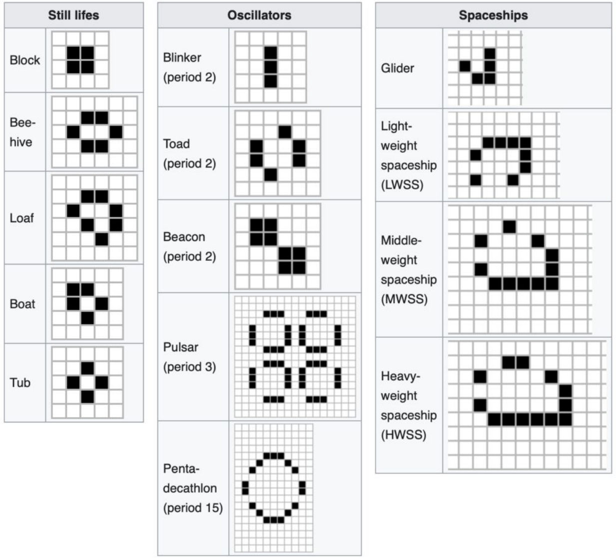
- Cellular Automata
    - Generalization of Conway’s Game of Life
    - Continuous states, space, and time
- “Lenia” (name of the organisms) are
    - Geometric
        - radial, bilateral, or rotational symmetry
    - Metametric
        - modular repetition of body parts or structures
    - Fuzzy
        - soft boundaries and values, not discrete
    - Resilient
        - retain their properties over time
    - Adaptive
        - change behavior in response to stimuli




Polymorphic Cells

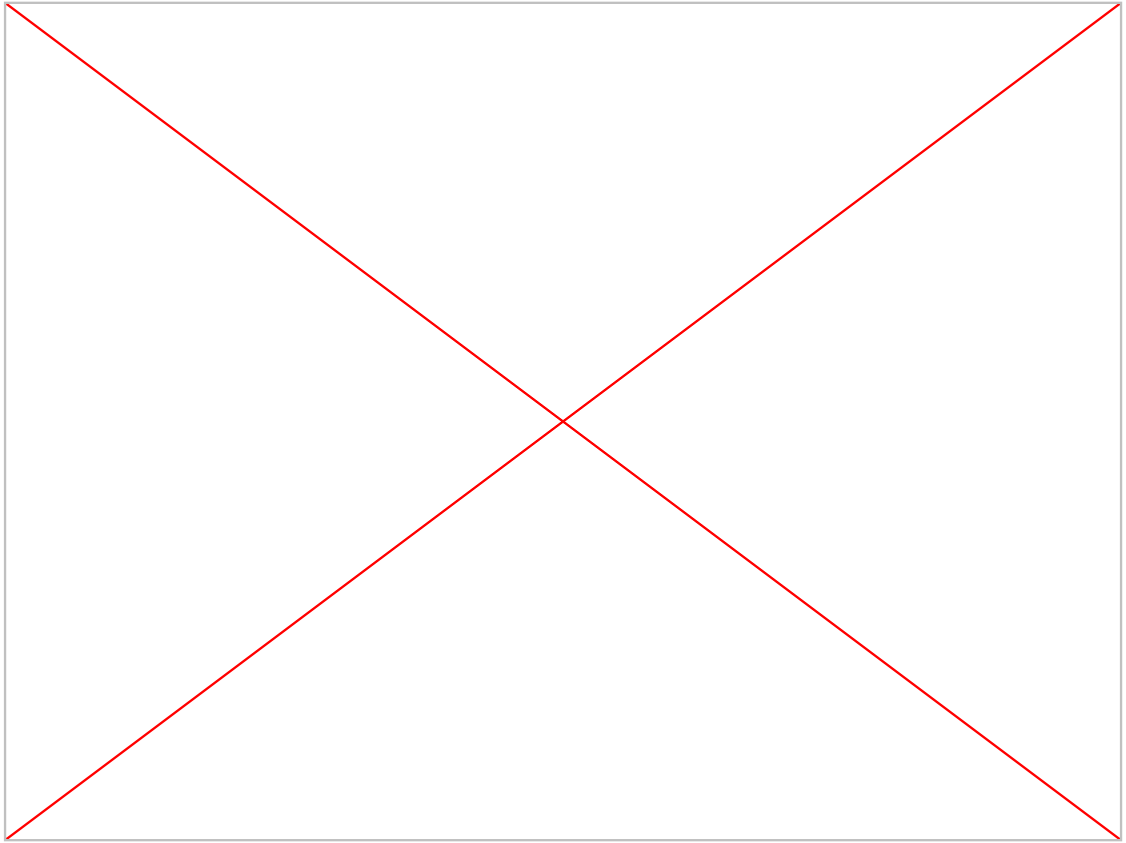
Will Greenlee - Lenia

- Inspirations
  - Conway’s Game of Life
  - Reaction-diffusion systems (e.g., Turing patterns).
  - The idea of digital creatures or proto-organisms that display autonomy, stability, and motility.
- Lenia is a superset (sort of)
  - Can approximate Conway’s Game of Life
    - R = 2
    - μ = 0.35
    - σ = 0.07
    - Smallest possible kernel

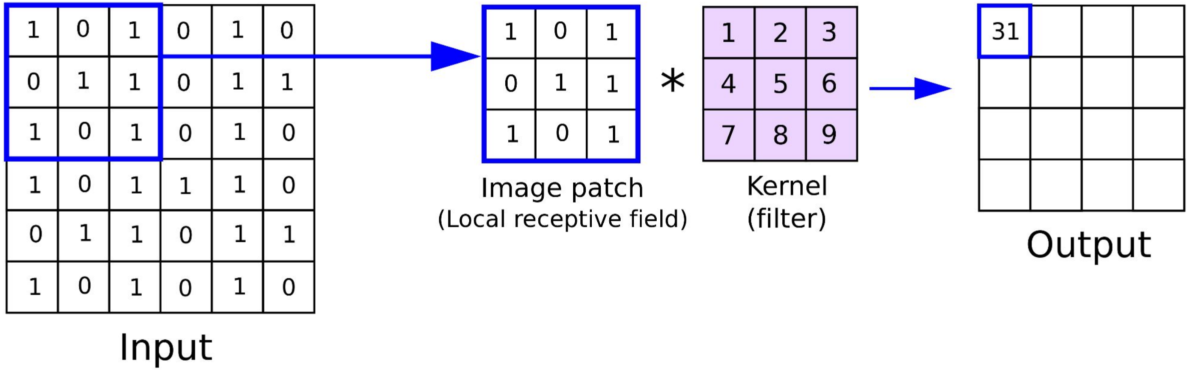


Will Greenlee - Lenia

- Experiment 1



- Lenia operates on:
  - A continuous grid
    - $A \subset R^n$
    - n is the dimension of the simulation, for this set of experiments it is 2
  - Real-valued state at each grid point
    - $u(x,t) \in [0,1]$
    - rendered as a heat-map of pixels
  - Time evolution is governed by three processes
    - Kernel Convolution, growth mapping, and state update rules (just like discrete CA)



Will Greenlee - Lenia

- Kernel convolution
  - The current state μ is convolved with a kernel K to compute the potential field U

$$U = K * u$$

- Growth mapping
  - μ = growth center, σ = growth width.

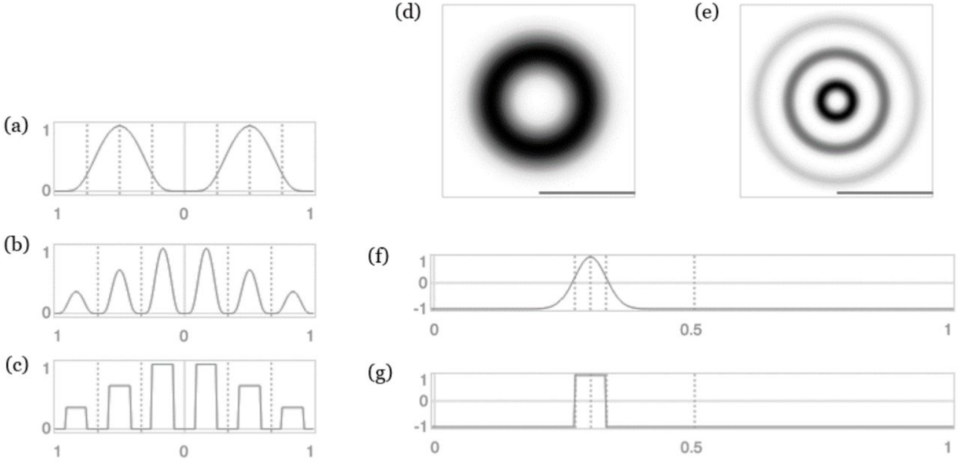
$$G(U) = 2 \cdot \exp\left(-\frac{(U-\mu)^2}{2\sigma^2}\right) - 1$$

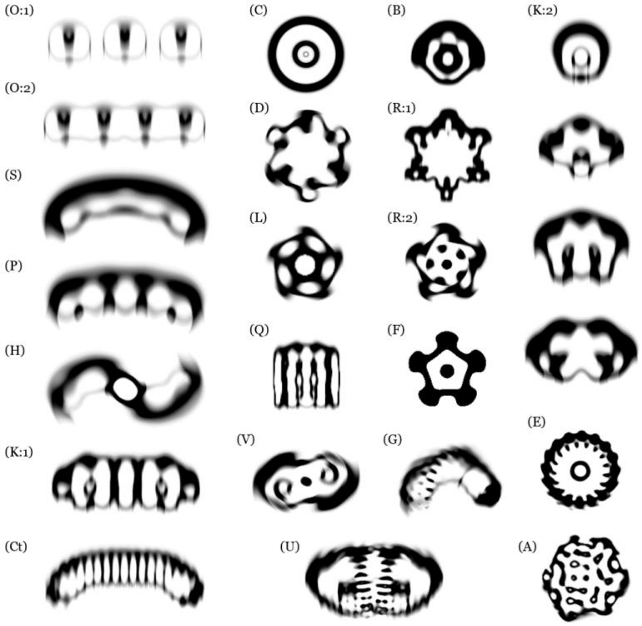
- Update rule
  - Δt: time step.
  - Clip keeps values in limits of u(x,t)∈[0,1]

$$u(t+1) = \mathrm{clip}(u(t) + \Delta t \cdot G(U), 0, 1)$$

Will Greenlee - Lenia

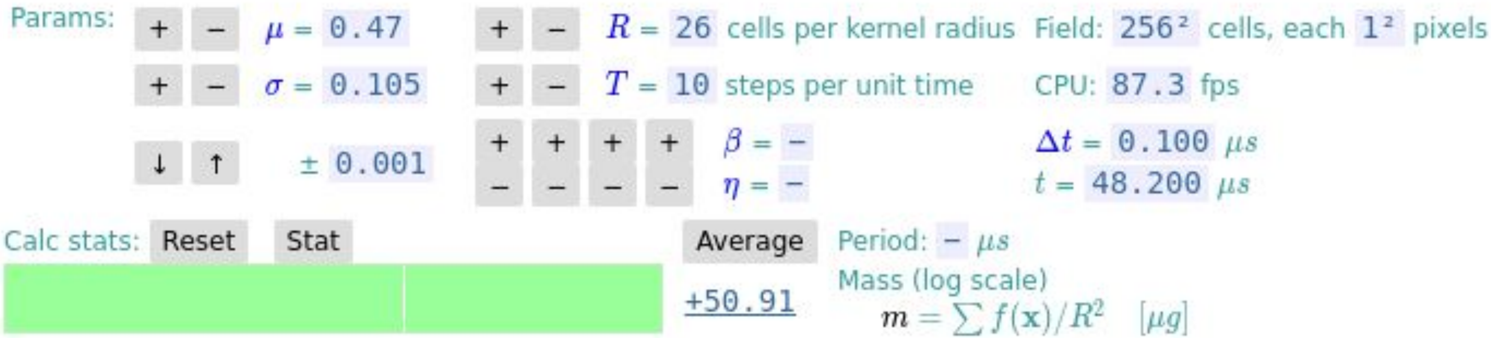
- Kernel Types
  - Exponential decay
  - Bell curve (Gaussian)
  - Multi-peak
- Kernel is nearly analogous to species





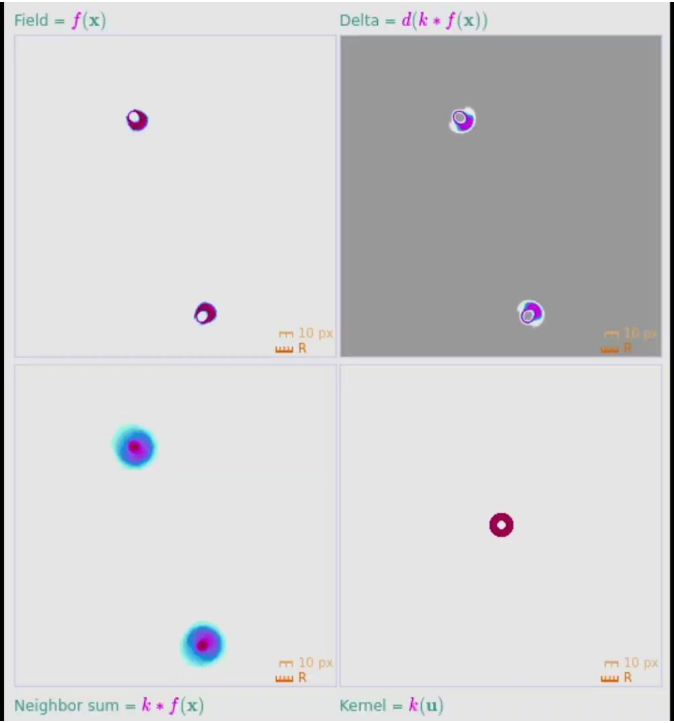
Will Greenlee - Lenia

- Key parameters:
  - μ: Growth center
  - σ: Growth width
  - R: Kernel radius
  - dt: Time resolution



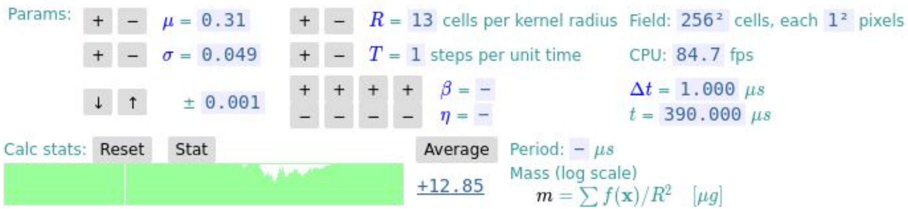
Will Greenlee - Lenia

- Experiment 2



Will Greenlee - Lenia

kernel



- Is it repeatable?
  - **Yes! (seeded randomness)**
- Key configuration features
  - The kernel and the growth parameters (μ = growth center, σ = growth width)
- Similar to the Edge of Chaos we discussed early in the quarter!
  - From randomness (the fat glider chaos interactions) | High entropy
  - Comes order (the regular gliders) | Low entropy
- DST
  - The normal gliders are attractors
    - The system tends toward this state
  - The fat gliders are semi-stable, and produce stable points chaotically
- Can be seen almost like a brain
  - Gliders are signals
  - Fat gliders are signal sources (not quite neurons though)
- Future Ideas
  - (If I could get the python running) design the fat gliders from the start
  - Try to create a more stable instance of the fat glider

Steven Johnson - GenePool

icon taken from the creator of the simulator’s website: https://www.ventrella.com/Alife/alife.html

Steven Johnson - GenePool

jeffrey ventrella

CV

ventrella.com

wikipedia

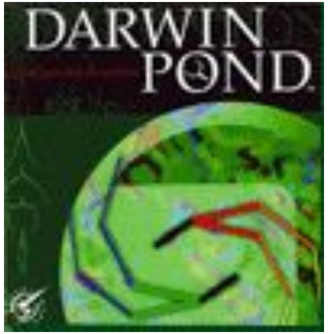
videos

I am an algorithmic artist, animated artificial life developer, interaction designer, virtual world builder, and visual language practitioner, working at the intersection of art, science, and math. I am antidisciplinarian by nature.

Steven Johnson - GenePool

History

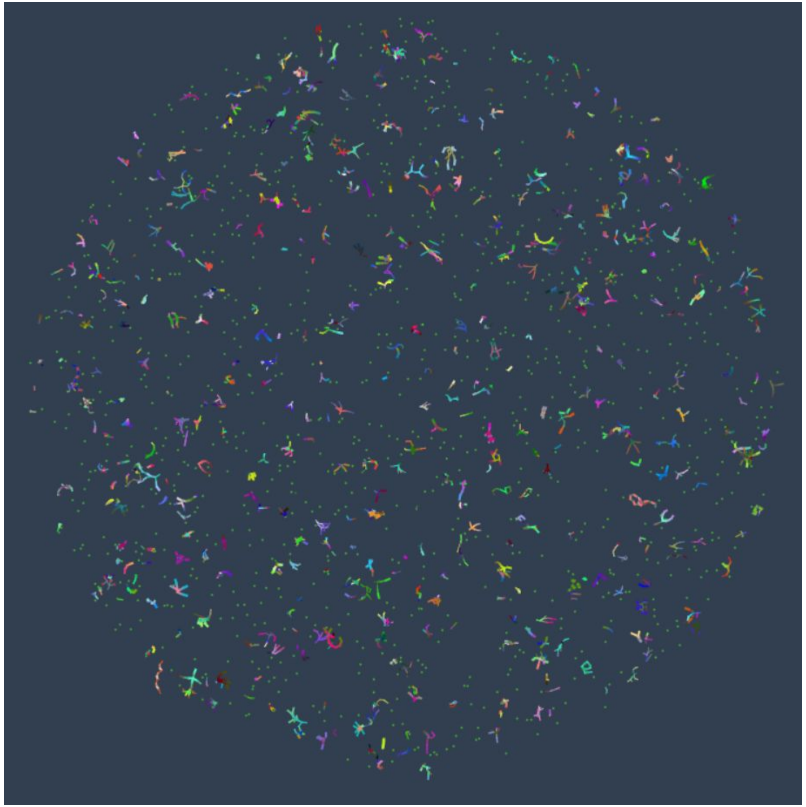
- 1996 - Darwin Pond
- “Darwin Pond is for nature lovers, who spend countless hours (or a lifetime) meditating on the wonders and intricacies of life unfolding.”
- “In Darwin Pond, hundreds of physically-based organisms achieve locomotion via genetically-based motor control and morphology. The ability to have more offspring is a direct outcome of two factors: 1) better ability to swim to within a critical distance to a chosen mate, and 2), the ability to attract other organisms who want to mate.”
- Rocket Science Games -> research project focus (‘96) -> game focus (‘97) -> studio goes down project, remains free for all users (‘98)



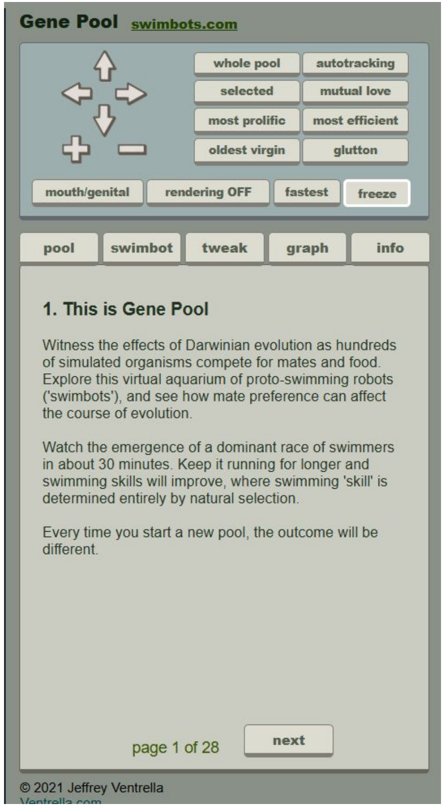
Steven Johnson - GenePool

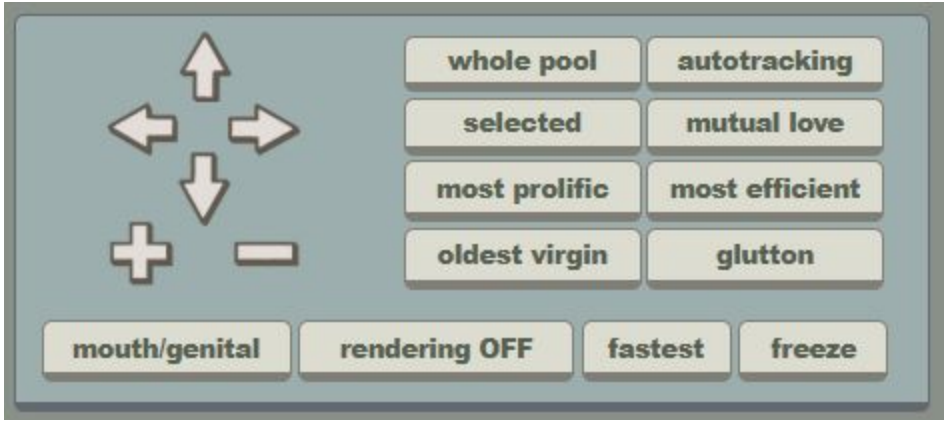
History

- 1998 - GenePool
  - Less interactive features (can't widget your own worms anymore)
  - Added more attractors (sex preferences) to experiment with
  - Ported to local client, website, and mobile eventually
  - Continually added upon and experimented with ever since
- 2021 - newest update (what I ran on)



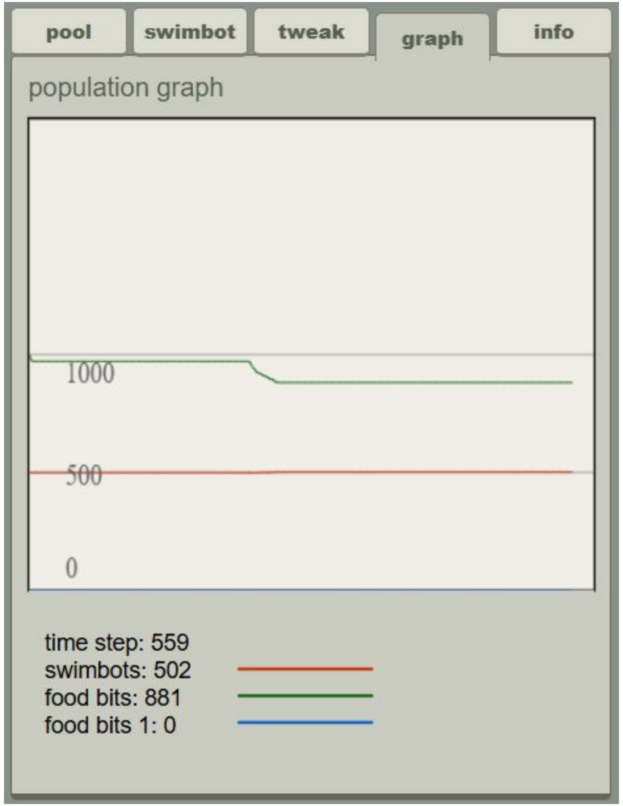
Steven Johnson - GenePool: Settings



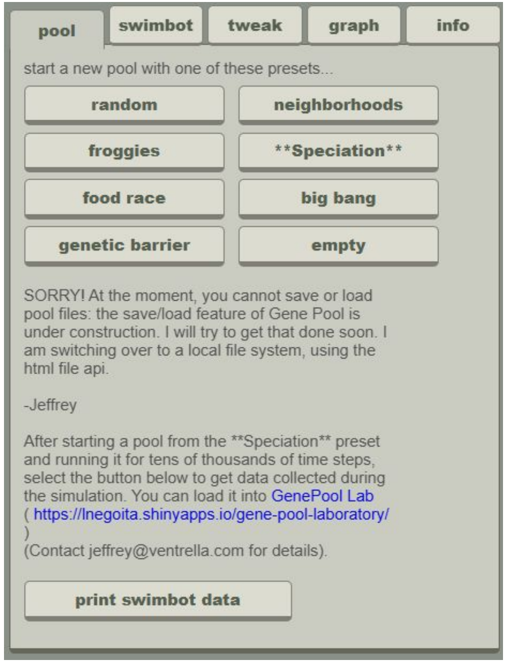


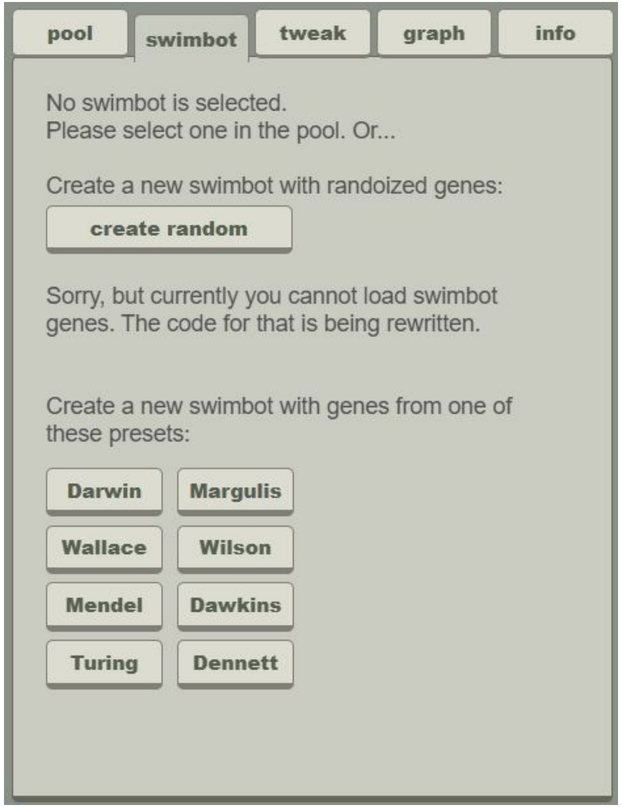
Viewer Panel

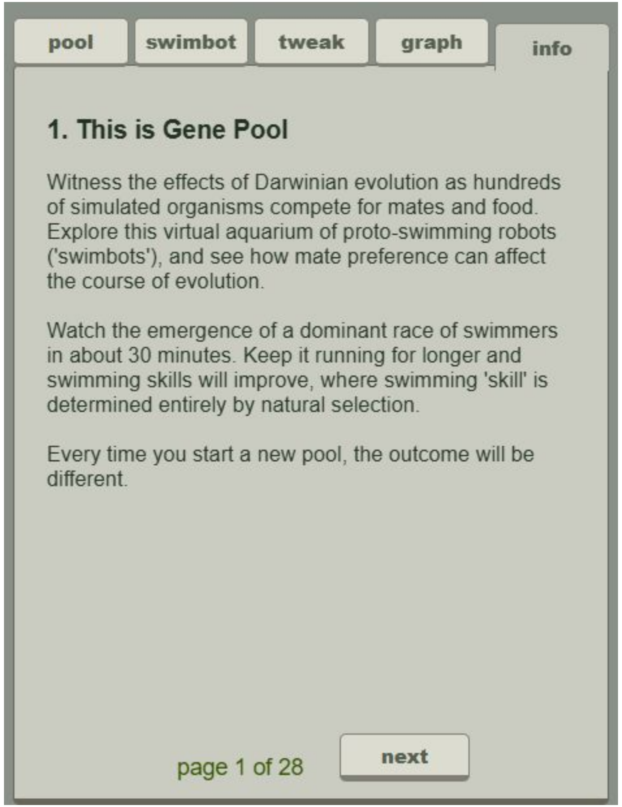
Steven Johnson - GenePool: Settings



Steven Johnson - GenePool: Settings







Steven Johnson - GenePool

Mechanisms/Flow

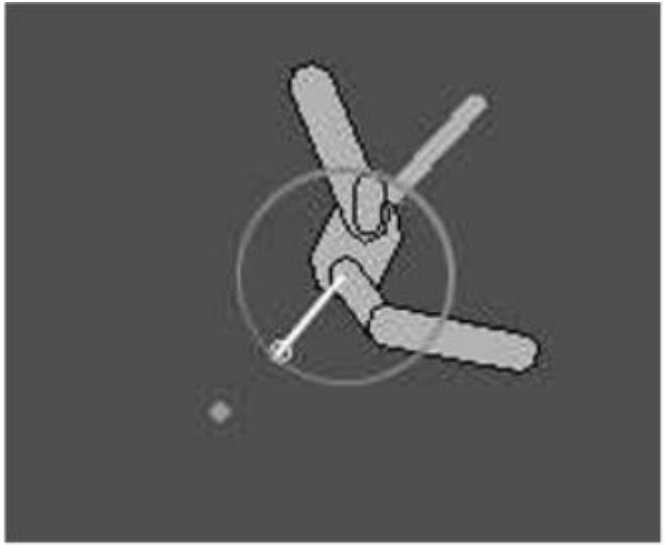


Figure 1.5. The mouth vector and a circle showing the critical distance for eating a food bit.

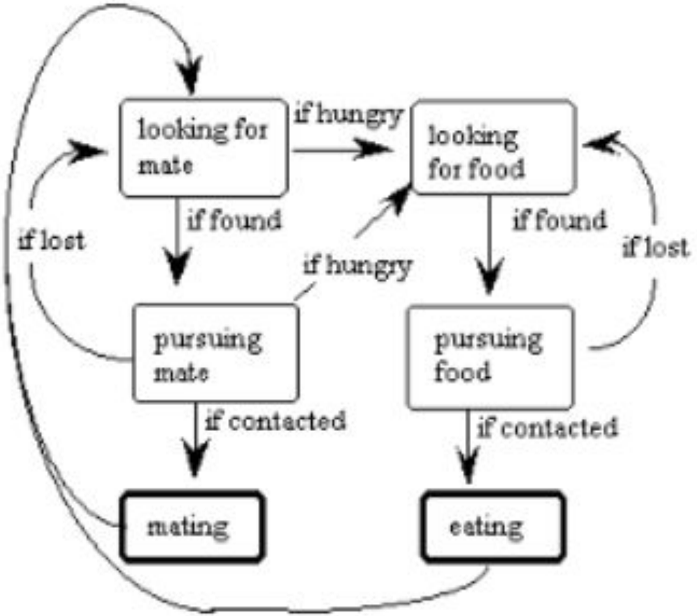


Figure 1.6. Swimbot mental states

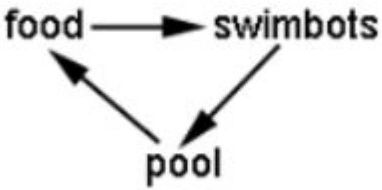


Figure 1.7. Energy flow in GenePool.

Steven Johnson - GenePool

Mechanisms/Flow

Figure 1.8. Swimbot orientation compared to goal direction modifies genetically-determined turning mechanisms.

Innate orientation based on axis of main body part.

When Goal Present:

- Direction from swimbot to goal is compared to its orientation repeatedly
- Size and sign of angle used to modify all part motions

The Questions

Main Questions Answered

- What is the simulator focused on?

- Exploring evolution in real-time in an easy-to-access way allowing the user to come up and test their own hypotheses.

- What levels does it attempt to capture?

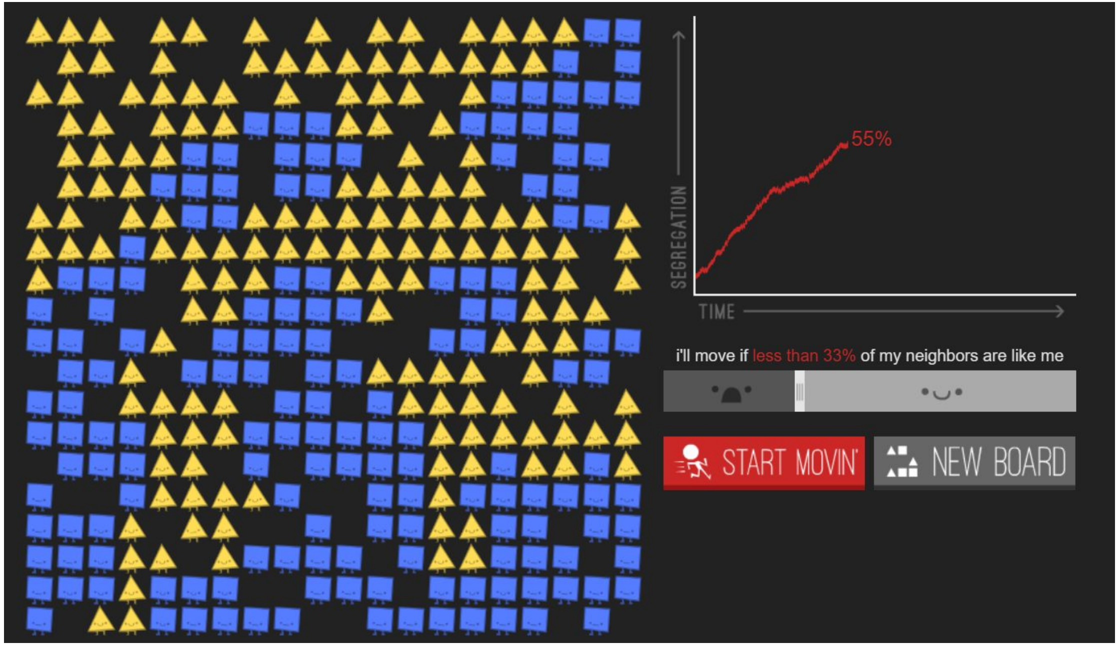
- Locomotion, aesthetics, space, food, and mating.
- “bring some basic principles of evolution to light”

- What is an example of a research question it purports to be able to help answer?

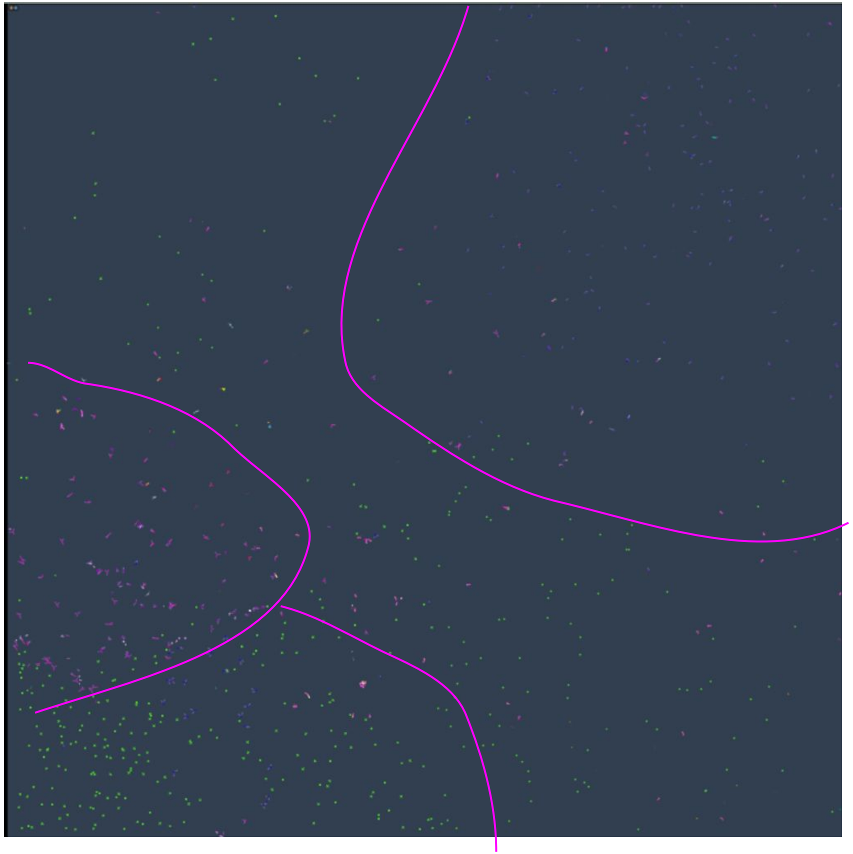
- Using the “still” attractor. Hypothesis: They go extinct.
  - What happened: sexual dimorphism
    - Majority: sexy “sitters” (they don’t move)
    - Minority: agile “breeders” (they have tail-whips allowing them to move around really fast)

My Experiment

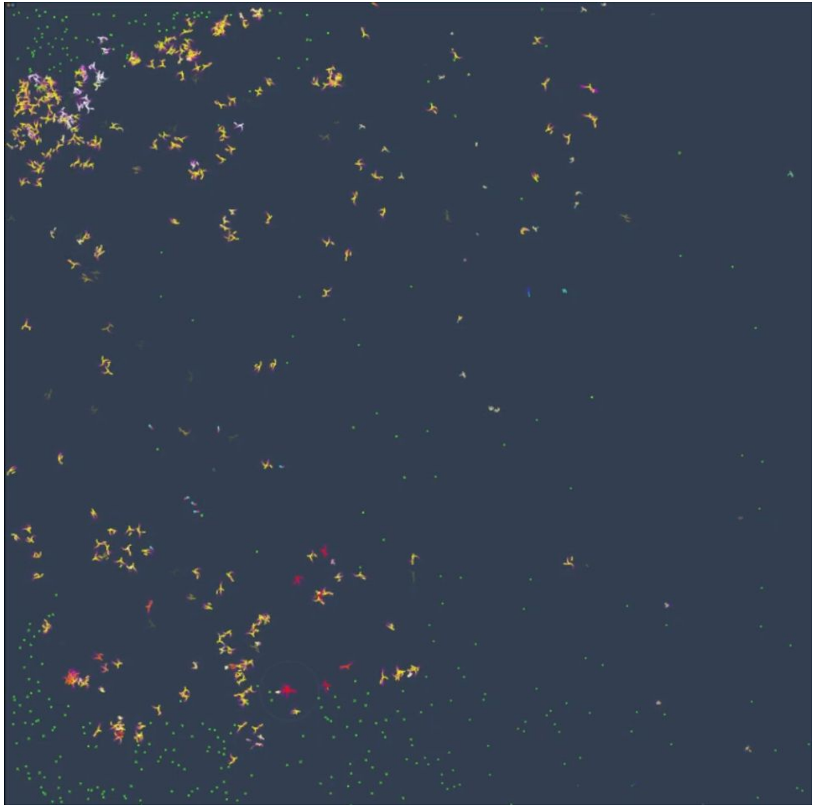
- Artificial Society inspiration
  - Polygon Segregation
- Question
  - Will the more advanced GenePool result in segregation like polygons did?
- Approach
  - Default settings (including “Similar color” attractor)

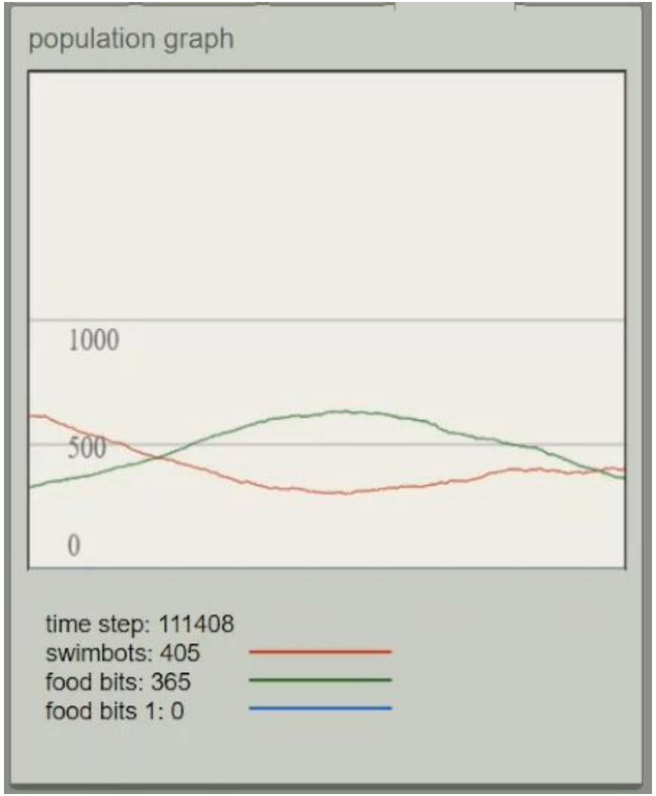


My Experiment: Results (Run 1 - no video)



My Experiment: Results (Run 2 - video)





Interesting results:

- Food geographic disparities
- One race emergence
- Mutations (red species) found in video

My Experiment: Required Questions

- Is it repeatable?
  - Yes.
- What are the most important mechanisms (change/remove components)?
  - Attraction rule (outside my experiment - access to food, no food = death)
- Can we understand in detail one specific instance of the observed results/behavior?
  - Similar color attractor leads to segregated populations just like the shapeism rule in polygons
- Could we possibly understand the system better through information theory?
  - Using entropy (randomness, uncertainty) and applying it to diversity. In this case, a higher entropy would mean higher diversity and we could then graph that over time to visualisize how diversity evolves throughout the simulation.
- Could we possibly understand the system better through dynamical systems theory?
  - Attractors being a mix between the attractor rules and the proximity to food
- Is there potential of seeing this behavior/phenomena in the real world?
  - How could you try to imitate this?
    - Gene editing
  - What would be crucial to consider in evaluating this?
    - Comparisons to simulations and insights gained from anthropology, archaeology, ecologist, and paleontology.

My Experiment: Cross-Reference and Wishes

Cross-Reference:

- Ventrella paper - Celebrating Diversity Experiment
  - “similar color” criteria
    - “swimbots will choose mates whose bodies contain the closest spectrum of colors to their own”
  - “encourage interracial mating by adding a new attractiveness criterion called “opposite color”. Not surprisingly, when this is turned on, the population converges on a perpetual state of psychedelic diversity.”

Wishes:

- I wanted to run opposite color attractor! (Why isn’t it an option!)
- Continual updates after 2021
  - Re-addition of saving and loading in newest update
- A better fast-forward/time slider!! (I had my laptop running the simulator for many hours and if I pulled up something else in full-screen it would stop!!)

Other Things I Did

- Random attractor
- Closest attractor
  - Observation: Took more than double the time as the others to get to similar timesteps
- Recorded videos for all three attractors I tested (hours and hours)
  - Could pull those up and look through them for added insight from the class! :)
- Kept the tabs open
  - If they haven't closed by the time we get into class we can continue running them and see what happens
  - Also, can print out the genetic data of each and compare it to the genetic data of a fresh start

Appendix

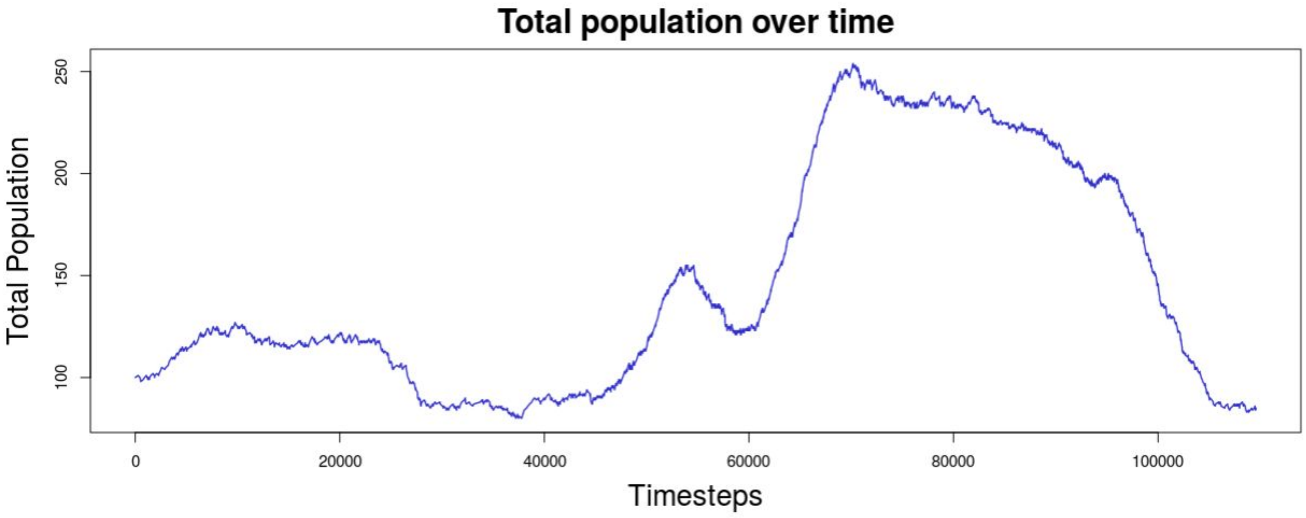
Simulation Summary:

5910
Total Swimbots

111363
Total Timesteps

40001
Maximum Lifespan

13598
Average Lifespan



Similar color (R1)

Appendix

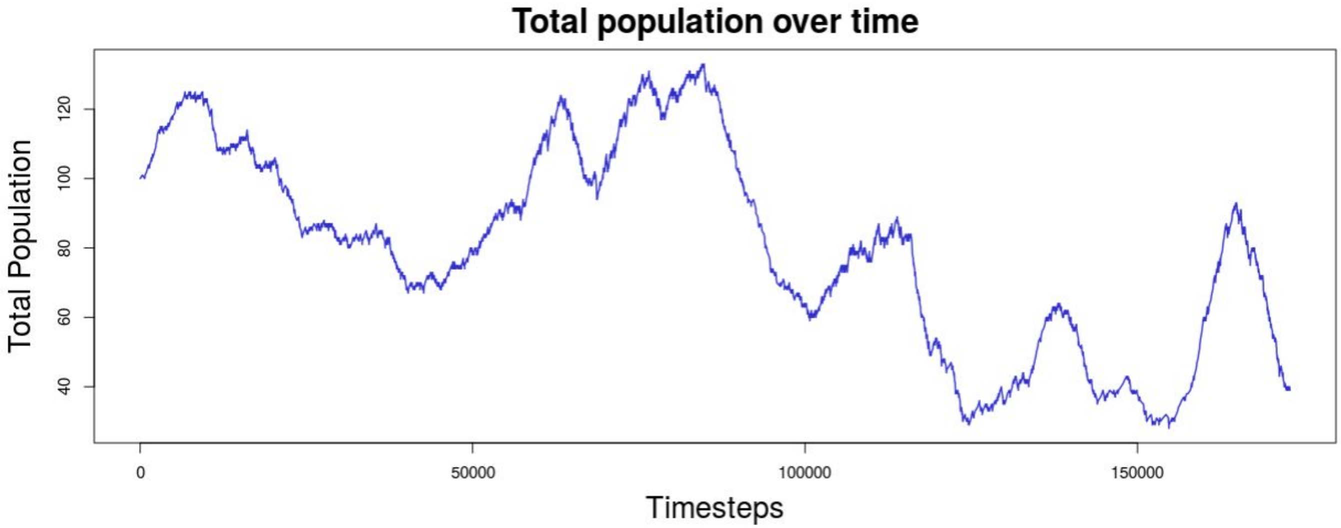
Simulation Summary:

6745
Total Swimbots

174878
Total Timesteps

40001
Maximum Lifespan

10358
Average Lifespan



Similar color (R2)

Appendix

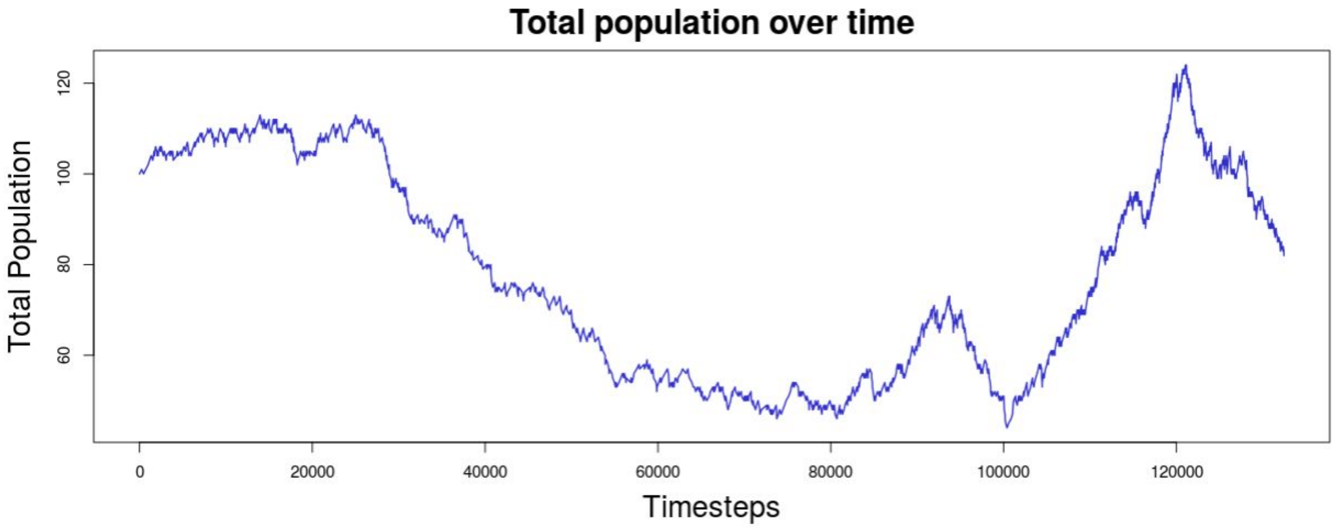
Simulation Summary:

4680
Total Swimbots

133296
Total Timesteps

40001
Maximum Lifespan

11144
Average Lifespan



Random

Appendix

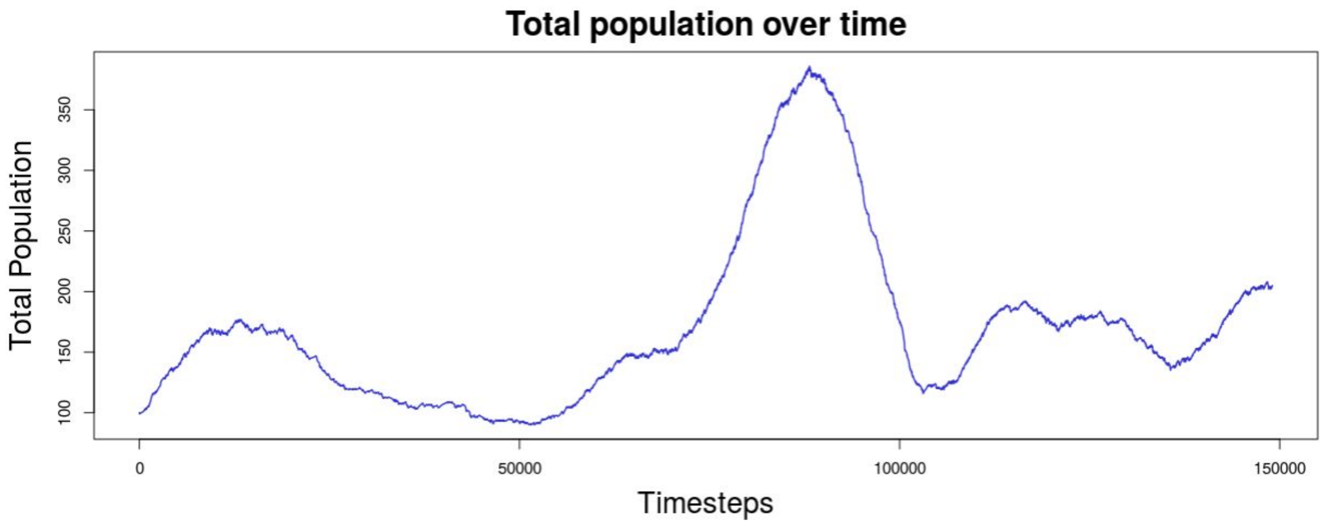
Simulation Summary:

8975
Total Swimbots

150788
Total Timesteps

40001
Maximum Lifespan

14269
Average Lifespan



Closest